

Bhargi Bhalara

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SUMMARY

An ardent engineer with over eight years of experience in integrating FPGA-based solutions into complex systems, showcasing expertise in RTL Designs for SoC/FPGAs/ASIC, VLSI, digital and analog circuit design, signal processing, and embedded systems. Demonstrated proficiency in designing advanced L1/L2 algorithms for wireless communication systems, particularly focusing on multi-user MIMO techniques. Skilled in end-to-end FPGA design flow with seamless post-design integration, validation, and debugging efforts. Proven ability to enhance system architecture with meticulous documentation and automating test processes, significantly improving efficiency and accuracy by up to 30%.

EXPERIENCE

Electrical Systems Engineer

Reflexion Medical, Inc.

Jun 2021 - May 2025

- Implemented end-to-end FPGA design flow including Synthesis, Implementation, Place and Route, and Bitstream Generation for Xilinx Ultrascale FPGAs using Vivado design suite, improving design cycle efficiency by 25%.
- Improvised the existing state machine to reduce the delay by 3-4 clock cycles, optimizing overall system performance and latency.
- Automated regression test cases using Python and standard verification methodologies, resulting in a 40% reduction in testing time and increased reliability of design validation.
- Developed embedded test programs and prepared validation protocols for testing bare metal systems, enhancing testing coverage by 50%.
- Debugged High-Speed Serial Interfaces like PCIe and CAN Bus using protocol analyzers, ensuring compliance with industry standards and reducing error rates by 20%.
- Initiated engineering change orders and supported the design transfer process for updated FPGA designs, facilitating smooth transitions and minimizing downtime.

Electrical Engineer

Mirion Technologies (Capintec) Inc.

Apr 2017 - Jun 2021

- Remodeled RTL design logic for FPGA/CPLD used in Multi-Channel Analyzers and related PCB layout, improving signal integrity and performance metrics by 15%.
- Supported design changes for electronic components like Microcontroller, SDRAM, PMIC, and similar CMOS devices in PCBs, ensuring adherence to ISO-13485 requirements and reducing time to production by 30%.
- Delivered design changes to production efficiently, following the design control process complying with regulatory bodies like IEC-60601 and EU Low Voltage Directive, achieving a 100% compliance rate.
- Led hardware validation for Dose Calibrators, Thyroid Uptake Systems, and Radiation Measurement Devices, ensuring accuracy and reliability in measurements, thereby increasing customer satisfaction by 20%.

Electronics Engineer

Reliance Electronics Inc.

Jan 2017 - Sep 2017

- Designed and validated printed circuit boards (PCB) using custom test fixtures, ensuring correct functionality post-manufacture and reducing error rates by 15%.
- Structured the BOM for circuit designs by selecting appropriate parts and updating designs due to EOL/obsolescence, ensuring continuity in production and minimizing delays by 20%.

Design Engineer

Scalable Systems Research Labs

Feb 2016 - Dec 2016

- Structured IEEE Floating Point Numbers Multiplier in Verilog for the Floating Point Unit used in RISC-V CPU, improving computational efficiency by 30%.
- Reduced delay on carry propagation by implementing vertical slice compression technique, enhancing processor speed by 25%.

FPGA Intern

Finisar Corporation

Jun 2016 - Sep 2016

- Developed comprehensive Verilog code for 10G Ethernet transceivers on Xilinx Kintex 7 FPGA, enabling high-speed data transmission with a throughput of 10Gbps.
- Enhanced power efficiency by optimizing the state machine for Microsemi FPGA, achieving a 15% reduction in power consumption.

Instructional Student Assistant

San Jose State University

Aug 2014 - Dec 2014

- Assisted students with their Advance Digital Logic Course to teach digital design concepts like combinational and sequential circuits, Logic Reduction Techniques, and simulation of VHDL code in ModelSim, contributing to improved student performance by 20%.

EDUCATION

Masters of Science in Electrical Engineering

San Jose State University, San Jose

Dec 2015

GPA: 3.4

Bachelor of Engineering

Sarvajani College of Engineering and Technology

May 2013

SKILLS

AMBA (AHB, APB, ASB, AXI), ASIC, C, C++, Chipscope/ModelSim/Questasim, JIRA, JTAG, Lab Equipments and tools, Linux, Mentor Graphics PADS, Shell, SOC, System Verilog, USB, Verilog, CPU Architecture, GIT, HDMI, Intel Quartus II/Prime, Jenkins, Microchip Libero SoC, PCIe/PCI, Python, QNX, QuestaSim, RTOS, VHDL, Vitis HLS, VUnit verification framework, Xilinx ISE Design Suite, Xilinx Vivado Design Suite